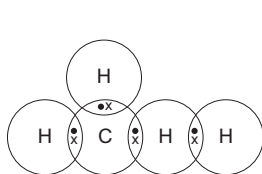
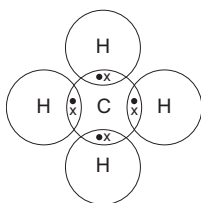
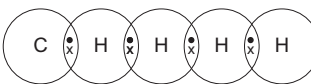
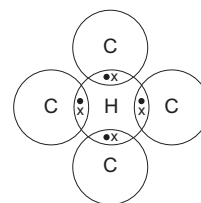


3. (a) Methane gas, CH_4 , contains the elements carbon and hydrogen.

Element	Electronic structure
carbon	2,4
hydrogen	1

- (i) I. Put a **tick** (✓) in the box below the diagram which shows how the atoms are bonded in one molecule of methane. [1]


☐

☐

☐

☐

- II. **Circle** the name given to the type of bonding found in methane. [1]

covalent

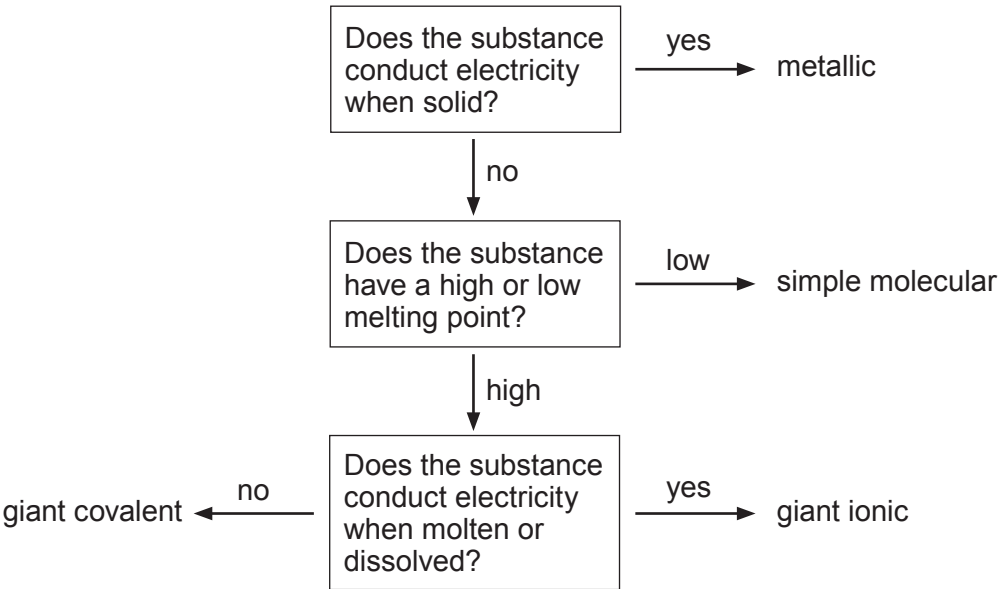
ionic

giant covalent

metallic



(ii) The flow chart below can be used to identify the type of structure found in different substances.



I. Use the flow chart to complete the table. [1]

Substance	Melting point (°C)	Electrical conductivity	Type of structure
A	2072	conducts only when molten	
B	-182	does not conduct electricity	
C	1610	does not conduct electricity	

II. Give the **letter** of the substance, **A**, **B** or **C**, that is most likely to be methane. [1]

Substance

3430U501
09



- (b) Butene, C_4H_8 , also contains the elements carbon and hydrogen. Calculate the percentage by mass of hydrogen in butene. [2]

$$A_r(H) = 1 \quad A_r(C) = 12$$

Percentage = %

Examiner
only

6

